

Deploy and Recover a Compute Engine Web VM

90 minutes · Mapped criteria A3, K4

Purpose and work product

Deploy a private web VM with a user-managed service account, test the startup script and recover from a bad change.

Submit: startup logs, a pre-change snapshot and recovery verification

Safety, cost and privacy

Use only the assigned training project. Confirm the active account, project and region before every mutation. Never place credentials, payment details, access tokens or private keys in screenshots or submitted files. Stop billable resources and verify the final inventory.

Files in this folder

File	Purpose
startup.sh	Mock data, script or configuration used by this activity
verify_vm.sh	Mock data, script or configuration used by this activity

Detailed procedure

Step 1

Review startup.sh and verify it is idempotent and contains no credentials.

Step 2

Create web-1 in asia-southeast1-b on app-sg without an external IP, attach app-runtime and add the web network tag.

Step 3

Supply startup.sh as instance metadata and wait until the VM is RUNNING.

Step 4

Run verify_vm.sh and inspect the serial output for package and service errors.

Step 5

Connect through IAP or an approved private path; do not create a public SSH rule.

Step 6

Run curl http://localhost and record the expected ACE activity healthy response.

Step 7

Create snapshot web-1-prechange before changing the web page or service configuration.

Step 8

Introduce the trainer-provided harmless configuration fault, restart Nginx and record systemctl plus journal output.

Step 9

Restore the known-good configuration or disk snapshot and repeat the curl test.

Step 10

Save the instance description, startup evidence, snapshot name and recovery result.

Step 11

Stop the VM when finished and verify Billing/Compute inventory.

Verification record

Record expected and observed output. A successful command without the correct project, identity, scope or evidence is not an accepted result.

Check	Expected evidence
Identity and scope	Active account, project ID, region or zone
Configuration	Resource description or policy output
Runtime result	Test output with UTC timestamp
Failure test	One controlled negative test and diagnosis
Cleanup	Final inventory and stopped/deleted billable resources

Troubleshooting sequence

Check the active identity and project first, then API enablement and IAM, then resource state, networking and quotas. Preserve the failed output before making a correction. If the issue remains, escalate with the exact command, timestamp, expected result, observed result and evidence files.

Cleanup

Delete or stop resources as instructed. Re-open Billing and the relevant service inventory, then record the remaining resource names. Budget alerts do not stop charges.